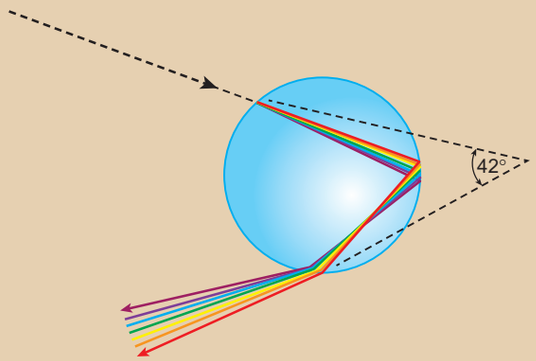
at different extents, inside the prism. *Moreover, refraction of a light ray is inversely proportional to its wavelength.*

Thus, the red coloured light, which has a large wavelength, is deviated less while the violet coloured light, which has a short wavelength, is deviated more.

DO YOU KNOW?

The formation of rainbow is an example of dispersion of white light. This can be seen on the opposite side of the Sun. After rainfall, large number of droplets still remain suspended in the air. When white light passes through them, it is split into seven colours. Dispersion of white light from a large number of droplets eventually forms a rainbow.




## Points to Remember

- Mirror is an optical device with a polished surface that reflects the light falling on it.
- Curved mirrors have surfaces that are spherical, cylindrical, parabolic and ellipsoid.
- If the curved mirror is a part of a sphere, then it is called a 'spherical mirror'.
- A spherical mirror, in which the reflection of light occurs at its concave surface, is called a concave mirror.
- A spherical mirror, in which the reflection of light occurs at its convex surface, is called a convex mirror.
- The focal length of a spherical mirror is half of its radius of curvature.
- Real images can be formed on a screen, while virtual images cannot be formed on a screen.
- Concave mirrors form a real image and it can be caught on a screen.
- Concave mirrors are used as make-up mirrors.
- Convex mirrors are used in vehicles as rear view mirrors.
- Based on the nature of the surface, reflection can be classified into two types namely, regular reflection and irregular reflection.
- The number of images formed by a mirror depends on the angle of inclination of the mirrors.

## A-Z GLOSSARY

| | |
|---|---|
| **Center of Curvature** | The center of the sphere from which the mirror is made. |
| **Dispersion of light** | Splitting of white light into its seven constituent colours (wavelength). |
| **Focal length** | Distance between the pole and the principal focus. |
| **Focus** | Point where the reflected rays converge at or appear to diverge from a point on the principal axis. |
| **Kaleidoscope** | Device which produces numerous and wonderful image patterns. |
| **Periscope** | Instrument used for viewing objects, which are over and around another body. |
| **Pole** | Point on the mirror's surface where the principal axis meets the mirror. |
| **Principal Axis** | Line joining the pole of the mirror and its center of curvature. |



8th_Science_Unit-3_EM.indd 32 03-03-2020 15:10:04